Yongdong Yeo

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Summary

I am a Ph.D. student advised by **Prof. Jung Hee Cheon** at the Department of Mathematical Science, Seoul National University. My research area includes various topics of **Homomorphic Encryption**, including its applications and related protocols.

Education

2019 - present	Ph.D. student at **Department of Mathematical Science**, **Seoul National University**. Advisor: Prof. Jung Hee Cheon	(GPA: 3.57/4.3)
(2022 - present	Research Area: Homomorphic Encryption and Cryptography)	
2014 - 2019	B.S. at **Department of Mathematics, Konkuk University**	(GPA: 4.22/4.5)

Experience

2026 Aug. - Sep.	Visiting Researcher at **SEFCOM Lab**, **Arizona State University**, Tempe, AZ, USA. Advisor: Prof. Gail-Joon Ahn
2026 Feb.	Visiting Researcher at **SEFCOM Lab**, **Arizona State University**, Tempe, AZ, USA. Advisor: Prof. Gail-Joon Ahn

Publications

Conference & Journal

Authors are listed in alphabetical order by last name, unless an asterisk (*) for the first author is indicated. A dagger (†) indicates the corresponding author, when applicable.

[1] Keewoo Lee* and **Yongdong Yeo***. "SophOMR: Improved Oblivious Message Retrieval from SIMD-Aware Homomorphic Compression". In: *35th USENIX Security Symposium (USENIX Security 26)*. Baltimore, MD: USENIX Association, Aug. 2026, pp. 3219–3238. ISBN: 978-1-939133-58-8. URL: https://www.usenix.org/conference/usenixsecurity26/presentation/lee-keewoo.

[2] Jung Hee Cheon, Hyeongmin Choe, Seunghong Kim, and **Yongdong Yeo**. "Multi-Party Homomorphic Encryption with Dynamicity and Ciphertext Reusability". In: *IACR Communications in Cryptology* 3.1 (May 4, 2026). ISSN: 3006-5496. DOI: 10.62056/a0qj89krz.

[3] Jihwan Kim*, Jung Hee Cheon, and **Yongdong Yeo**†. "OverModRaise: Reducing Modulus Consumption of CKKS Bootstrapping". In: *IACR Communications in Cryptology* 2.3 (Oct. 6, 2025). ISSN: 3006-5496. DOI: 10.62056/a3n5qjp10.

[4] Jung Hee Cheon, Minsik Kang, Taeseong Kim, Junyoung Jung, and **Yongdong Yeo**. "Batch Inference on Deep Convolutional Neural Networks With Fully Homomorphic Encryption Using Channel-By-Channel Convolutions". In: *IEEE Transactions on Dependable and Secure Computing* 22.2 (2025), pp. 1674–1685. DOI: 10.1109/TDSC.2024.3448406.

Preprints

[1] Jung Hee Cheon, Keewoo Lee, Jai Hyun Park, and Yongdong Yeo. *SIMD-Aware Homomorphic Compression and Application to Private Database Query*. 2024. arXiv: 2408.17063 [cs.CR]. URL: https://arxiv.org/abs/2408.17063.

Honors & Awards

(Korea) National Cryptography Contest

2025,	Encouragement Award, $1500	SophOMR: Improved Oblivious Message Retrieval from SIMD-Aware Homomorphic Compression
2024,	Encouragement Award, $1500	Reusable Dynamic Multi-Party Homomorphic Encryption
2024,	Special Award, $500	OverModRaise: Reducing Modulus Consumption of CKKS Bootstrapping
2023,	Special Award, $500	Private Database Queries with SIMD-Aware Homomorphic Compression
2023,	Special Award, $500	Batch Inference on Deep Convolutional Neural Networks with Fully Homomorphic Encryption Using Channel-By-Channel Convolutions

Invited Talks & Presentations

2026 Aug.,	USENIX Security 2026:	SophOMR: Improved oblivious message retrieval from SIMD-aware homomorphic compression
2026 Jun.,	PACOH KIAS Workshop:	SophOMR: Improved oblivious message retrieval from SIMD-aware homomorphic compression
2025 spring,	KMS Spring Meeting:	SophOMR: Improved oblivious message retrieval from SIMD-aware homomorphic compression
2024 fall,	KMS Annual Meeting:	Reusable Dynamic Multi-Party Homomorphic Encryption
2024 Aug.,	MPC & SNARK Workshop:	SophOMR: Improved oblivious message retrieval from SIMD-aware homomorphic compression
2023 Dec.,	Dept. of Math at Konkuk University	Homomorphic Encryption and Its Application to Machine Learning
2023 fall,	KMS Annual Meeting:	Private database query with SIMD-aware homomorphic compression

Teaching

Teaching Assistant, Seoul National University

Duties include grading, office hours, and exam administration. Courses marked with † were taught as a recitation instructor, leading weekly recitation sessions.

2026 Fall,	Calculus Practice 2†
2026 Spring,	Modern Algebra 1
2025 Spring,	Mathematics for Business Administration
2024 Fall,	Mathematics for Life Sciences 2
2024 Spring,	Undergraduate Thesis Seminar in Mathematical Sciences
2023 Fall,	Cryptography
2023 Spring,	Calculus Practice 1†
2022 Fall,	Calculus Practice 2†, Theory of Information Security
2022 Spring,	Calculus Practice 1†
2021 Fall,	Calculus Practice 2†, Undergraduate Thesis Seminar in Mathematical Sciences
2021 Spring,	Calculus Practice 1†, Modern Algebra 1
2020 Fall,	Honor Calculus Practice 2†, Linear Algebra 2
2020 Spring,	Honor Calculus Practice 1†
2019 Fall,	Calculus Practice 2†

Languages

Korean	Native language.
English	2nd language. Fluent.
Japanese	3rd language. Conversational.
Programming language:	C++, Go, Python, MatLab. Experience with AI-assisted tools.